

VALIDRA

Problem Statement ID	26034
Problem Statement Title	Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels
Theme	Miscellaneous
PS Category	Software
Team Name	VisionMinds



THE PROBLEM WE SOLVE

- Every packaged commodity must carry mandatory declarations - manufacturer/packer, net quantity, MRP, dates, consumer-care.
- Manual enforcement is slow and unscalable across millions of SKUs in retail and e-commerce.
- Common violations: missing declarations, wrong MRP format, undersized fonts, improper placement.

OUR SOLUTION - VALIDRA

AI extracts > rules decide > RAG explains > humans review

- 1 **SCAN** Upload or capture label images; originals kept as tamper-evident evidence.
- 2 **EXTRACT** OpenCV preprocessing + OCR with bounding boxes localise every declaration.
- 3 **VALIDATE** Versioned rule engine checks presence, correctness, format, placement, font size.
- 4 **EXPLAIN** RAG over the Act & Rules attaches the exact legal provision to each finding.
- 5 **REPORT** PDF + editable reports with cropped evidence, confidence and rule references.
- 6 **REVIEW** Searchable inspection history and enforcement dashboard with RBAC.

UNIQUENESS & INNOVATION

- **Separation of decision & language** - Rule engine decides; LLM only explains - no hallucinated legal verdicts.
- **Confidence-aware pipeline** - OCR > extraction > validation confidence propagated; low = NEEDS REVIEW.
- **Legal RAG with citations** - Every finding carries the retrieved provision of the Act / Rules.
- **Versioned rule repository** - Rules carry effective dates; packs judged by law at inspection date.
- **Evidence-grade audit trail** - Immutable records, bounding-box evidence and audit logs.

DECISION STATES

COMPLIANT

Declaration present and passes validation at high confidence.

NON-COMPLIANT

Declaration missing or invalid; severity classified and cited.

NEEDS REVIEW

Extraction confidence below threshold; routed to a human officer.

COMPLIANCE CHECKS PERFORMED



BEYOND THE PROBLEM STATEMENT

- **E-commerce listing crawler** - validate online product pages at scale.
- **Bulk batch mode** - inspect an entire consignment in a single job.
- **Consumer-facing check** - shoppers verify packs and report violations.
- **Rule-as-config** - amendments ship as new rule versions, no redeploy.

OPEN SOURCE

Zero per-scan licence cost

EXPLAINABLE

Every verdict cites the law

EVIDENCE-GRADE

Immutable audit trail

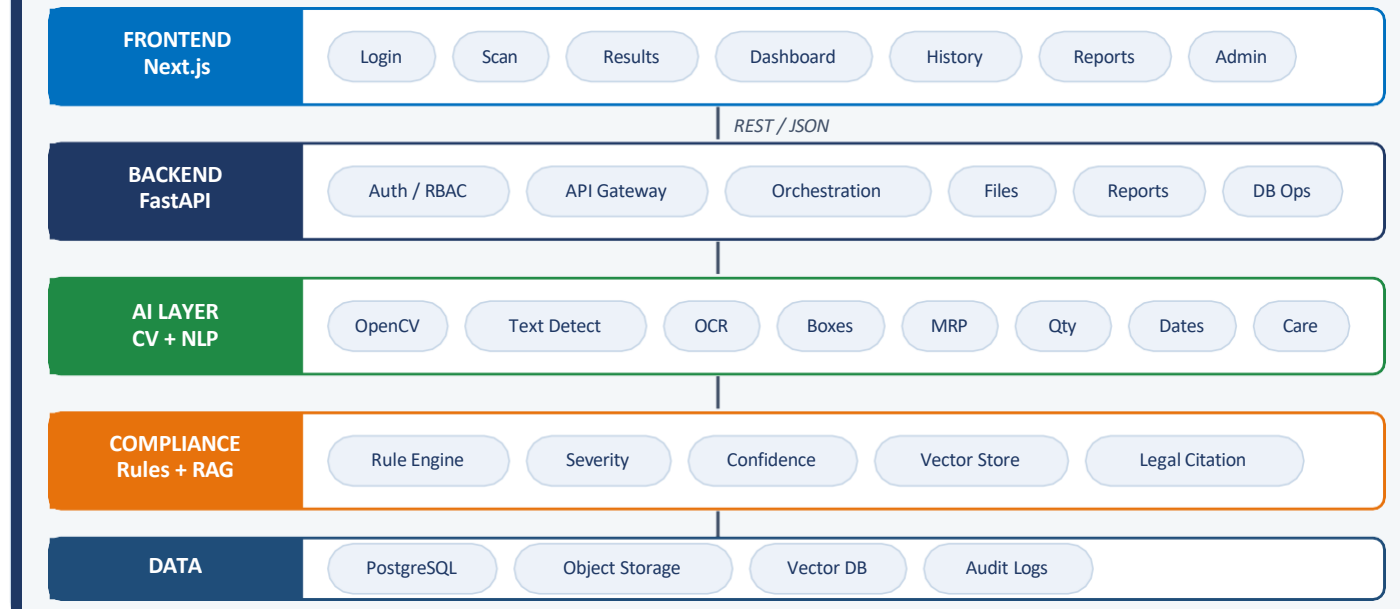
FIELD-READY

Web + mobile capture

PROCESSING PIPELINE



SYSTEM DESIGN / ARCHITECTURE



TECHNOLOGY STACK

Frontend	Next.js · TypeScript · Tailwind · shadcn/ui
Backend	FastAPI · Python · async REST · OpenAPI
Computer Vision	OpenCV - preprocessing, perspective fix
OCR	PaddleOCR - text, boxes, token confidence
ML / NLP	spaCy · scikit-learn - field extraction
Rule Engine	Versioned repository with legal references
RAG	Vector store over Act & Rules + LLM explainer
Database	PostgreSQL - inspections, rules, audit logs
Reports	ReportLab - PDF compliance reports
Security	JWT auth · role-based access · audit trail

IMPLEMENTATION METHODOLOGY - 6 PHASES



FEASIBILITY OF THE IDEA

TECHNICAL FEASIBILITY

- Built on mature, production-proven open source: OpenCV, PaddleOCR, spaCy, FastAPI, PostgreSQL, Next.js.
- No novel model research needed - innovation sits in the rule engine, calibration and evidence layer.
- Font-size checks reduce to measurable geometry once pixel-to-mm calibration is fixed.

OPERATIONAL FEASIBILITY

- Mirrors the officer's workflow: inspect > record > report. No behavioural re-training.
- Web-first with mobile capture - runs on hardware field officers already carry.
- Rules are configuration, so departments update them without engineering effort.

ECONOMIC & LEGAL FEASIBILITY

- Open-source stack, on-premise deployment: no per-scan licence cost, full data sovereignty.
- Positioned as decision support - final legal judgement stays with the authorised officer.
- Audit logs and immutable evidence align with enforcement record-keeping requirements.

CHALLENGES & MITIGATION STRATEGIES

! OCR degradation on real packs

RISK Glare, curved surfaces, reflective foil and multilingual typography reduce OCR accuracy.

MITIGATION Multi-stage defence: adaptive preprocessing (deskew, perspective correction, glare suppression), multi-engine OCR with agreement voting, confidence gate routing weak reads to human review.

! Font-size measurement in mm

RISK Rules prescribe physical sizes, but images only give pixels.

MITIGATION Reference-based calibration: derive scale from known references such as barcode module width, report with a tolerance band, flag borderline cases as NEEDS REVIEW.

! Legal ambiguity and amendments

RISK Rules are category-specific, exemption-heavy and periodically amended.

MITIGATION Versioned rule repository: every rule carries code, category scope, severity, legal reference and effective date. Amendments ship as new rule versions.

! Trust & admissibility of AI output

RISK Enforcement action cannot rest on an opaque model verdict.

MITIGATION Explainability by construction: the LLM never decides, it only cites. Each finding exposes extracted value, bounding-box evidence, confidence score and exact provision.

TARGET AUDIENCE & IMPACT

LEGAL METROLOGY ENFORCEMENT

- First-pass inspection automated - officer time shifts to acting on flagged exceptions.
- Consistent, repeatable verdicts across officers, districts and product categories.
- Evidence-backed reports generated at inspection time, not reconstructed afterwards.

CONSUMERS

- Faster detection of missing MRP, wrong net quantity and unreadable declarations.
- Stronger protection against unfair trade practice and misleading packaging.
- Optional consumer check-and-report channel feeds field intelligence back.

INDUSTRY & E-COMMERCE

- Manufacturers and platforms can self-audit artwork before print or listing.
- Fewer recalls, seizures and penalties through pre-emptive compliance checks.
- A level playing field for compliant brands competing against non-compliant sellers.

BENEFIT ANALYSIS

SOCIAL BENEFITS

- Transparent, verifiable pricing and quantity at the point of sale.
- Greater consumer trust in packaged goods and in enforcement itself.
- Accessible reporting lets citizens join market surveillance.

ECONOMIC BENEFITS

- Sharp reduction in inspection cost per product versus manual checking.
- Higher enforcement throughput without proportional headcount growth.
- Reduced litigation risk - findings arrive with evidence and citations.

ENVIRONMENTAL & SYSTEMIC BENEFITS

- Paperless inspections replace printed forms and physical files.
- Fewer reprints and recalls of non-compliant packaging artwork.
- Nationwide analytics reveal violation trends and repeat offenders.

EXPECTED IMPACT INDICATORS

90%+

less manual label-reading time

100%

evidence preserved, immutable trail

1 verdict

consistent across all districts

Real-time

compliance trend analytics

Zero

per-scan licence cost

STATUTE & OFFICIAL SOURCES

- The Legal Metrology Act, 2009 - Ministry of Consumer Affairs, Food & Public Distribution (consumeraffairs.nic.in)
- The Legal Metrology (Packaged Commodities) Rules, 2011, as amended (consumeraffairs.nic.in)
- SIH 2026 - Problem Statement 26034, Department of Consumer Affairs (sih.gov.in/sih2026PS)
- Legal Metrology Division circulars & notifications (consumeraffairs.nic.in)

COMPUTER VISION & OCR

- PaddleOCR: Practical Ultra-Lightweight OCR System - Baidu Research (github.com/PaddlePaddle/PaddleOCR)
- EAST: Efficient & Accurate Scene Text Detector - CVPR 2017 (arxiv.org/abs/1704.03155)
- TrOCR: Transformer-based OCR with Pre-trained Models - AAAI 2023 (arxiv.org/abs/2109.10282)
- OpenCV documentation - preprocessing & perspective transform (docs.opencv.org)

RETRIEVAL-AUGMENTED GENERATION & LEGAL NLP

- RAG for Knowledge-Intensive NLP Tasks - Lewis et al., NeurIPS 2020 (arxiv.org/abs/2005.11401)
- Dense Passage Retrieval for Open-Domain QA - EMNLP 2020 (arxiv.org/abs/2004.04906)
- LegalBench: Benchmark for Legal Reasoning in LLMs - NeurIPS 2023 (arxiv.org/abs/2308.11462)
- Sentence-BERT: Siamese BERT-Networks - EMNLP 2019 (arxiv.org/abs/1908.10084)

PROJECT REPOSITORY & DOCUMENTATION

- VALIDRA - source code, architecture and technical documentation (github.com/dhruvpatel16120/Validra)
- System architecture, data model and API contracts (github.com/dhruvpatel16120/Validra/tree/main/docs)
- FastAPI & Next.js official documentation (fastapi.tiangolo.com | nextjs.org/docs)
- spaCy - industrial-strength NLP for structured field extraction (spacy.io)

KEY RESEARCH TAKEAWAYS

- Rule-based verdicts outperform end-to-end LLMs on statutory checks.
- Confidence gating is essential to keep OCR error out of decisions.
- Citation-linked findings are what make AI output admissible.
- Versioned rules absorb amendments without model retraining.

VALIDATION DATASETS & BENCHMARKS

- Curated pack-label corpus across food, cosmetics and household categories.
- Ground-truth annotations for MRP, net quantity, dates and consumer-care fields.
- Metrics: field-level extraction F1, rule-decision accuracy, review-routing precision.
- Adversarial set: glare, curved packs, low-contrast and multilingual labels.

STANDARDS & COMPLIANCE PRACTICE

- Declarations, font-size and placement norms per PC Rules, 2011.
- Evidence handling aligned with enforcement record-keeping practice.
- Role-based access and audit logging for admissible inspection records.
- Versioned rules mapped to amendment effective dates.